

MARK IVAN UGALINO

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RESEARCH INTERESTS

I run large-scale simulations of astrophysical objects and run analysis pipelines that transform these data products into synthetic observables that can be compared to real-world measurements.

1. Galaxy simulations, magnetic fields in galaxies, star formation
2. White dwarfs, supernova simulations, magnetic fields in (merger of) white dwarfs
3. Code development, numerical algorithms for solving PDEs
4. Astronomy and physics education

EDUCATION

University of Maryland, College Park Fall 2023 – Spring 2028 (*expected*)
Doctor of Philosophy in Astronomy, *PhD candidate since Fall 2025*
Department of Astronomy
Thesis: Galactic magnetic fields from star-forming clouds to the cosmic web

University of Massachusetts, Dartmouth 2021 – 2023
Master of Science in Physics
College of Engineering
Thesis: Turbulently driven deflagration-to-detonation transition in near- M_{ch} white dwarfs

University of the Philippines, Diliman 2018 – 2020
Master of Science in Physics
National Institute of Physics
Thesis: Dynamical friction effects on circular orbits immersed in a finite gaseous background

University of the Philippines, Diliman 2013 – 2018
Bachelor of Science in Physics
National Institute of Physics
Thesis: Density perturbation induced by relativistic bodies in slightly eccentric orbits

Quezon City Science High School 2009 – 2013
High school diploma

MENTORING

Nar Jasen Dela Cruz (BS Physics '26, University of the Philippines Baguio, co-adviser
now Academia Sinica Institute of Astronomy and Astrophysics summer student)
Project: *Investigating lunar formation scenarios using SPH*

Jin Young Kim (Wake Technical Community College *through* GRAD-MAP Summer Scholars 2025,
now BS Physics, University of North Carolina Chapel Hill)
Project: *Magnetic fields in galaxies*

Emau Argueta (BS Mathematics, University of Maryland)

Ana Fernandez-Napp (Prince Georges Community College *through* GRAD-MAP Summer Scholars 2026)

Prabodha Mudalige (MS Physics, UMass Dartmouth *now* PhD Physics, UT Knoxville)
Project: *Viscous evolution of astrophysical disks*

Vrutant Mehta (MS Physics, UMass Dartmouth *now* PhD EAS, UMass Dartmouth)

Jack Sullivan (BS Physics, UMass Dartmouth *now* PhD Physics, University of Connecticut)

PEER-REVIEWED PUBLICATIONS

* - Major contributions as co-author, † - Minor contributions as co-author, **Mentored student**

5. (3.) *Towards robust simulations of the galactic dynamo: a cross-code comparison between Eulerian and Lagrangian schemes*

MI Ugalino, B Diemer, V Semenov, **J.Y. Kim**, F Marinacci, C Pfrommer, M Schaller, N Shchutskyi, V Springel, R Teyssier, O Zier (2026) *ApJ* (in preparation)

4.* *First-Principles Turbulence-Driven Deflagration-to-Detonation Transition Mechanism for Near-Chandrasekhar Mass White Dwarf Progenitors*

K Patel, R Fisher et al, incl **MI Ugalino** (2026) *ApJ* (in press, arXiv)

3.* *Using ^{44}Ti Emission to Differentiate Between Thermonuclear Supernova Progenitors*

D Kosakowski, **MI Ugalino**, R Fisher, et al (2023), *MNRAS Letters* 519(1), L74–L78

2. (2.) *Steady-state density perturbations induced by a point mass in a finite cylinder.*

MI Ugalino, MFI Vega (2020) *Proceedings of the 38th Samahang Pisika ng Pilipinas Physics Congress*

1. (1.) *Density perturbations in a collisional fluid induced by a particle on a slightly-eccentric orbit.*

MI Ugalino, MFI Vega (2018) *Proceedings of the 36th Samahang Pisika ng Pilipinas Physics Congress*

THESES

3. **MS Thesis** (UMass Dartmouth) - Turbulently driven deflagration-to-detonation transition in near-Chandrasekhar white dwarfs: a thesis in Physics (PDF)

2. **MS Thesis** (University of the Philippines Diliman) - Dynamical friction effects on circular orbits immersed in a finite gaseous background (PDF)

1. **BS Thesis** (University of the Philippines Diliman) - Density perturbation induced by relativistic bodies in slightly-eccentric orbits (PDF)

GRANTS AND RECOGNITIONS

Do Good Campus Fund Grant (USD 6,000.00) 2026
“GRAD-MAP: A unique model for graduate student-led mentoring and outreach in astronomy and physics”, (with Yash Gursahani, lead)

Jacob K. Goldhaber Travel Grant (USD 600.00) Spring 2026
Awarded by UMD Graduate School for participation in “RAMSES User Meeting (RUM) 2026”

Maryland Space Grant Consortium Activity Grant (USD 28,340.00) 2026
“GRAD-MAP: A unique model for graduate student-led mentoring and outreach in astronomy and physics”, (Co-PI, with Yash Gursahani and Benedikt Diemer)

Jacob K. Goldhaber Travel Grant (USD 250.00) Fall 2025
Awarded by UMD Graduate School for participation in “Magnetic Fields and Cosmic Rays across Diverse Scales: What’s Next”

2023 NASA FINESST Research Grant (USD 148,826.00) declined
Among first 27 selected for award out of 262 submitted proposals (10.3% award rate)

Graduate Dean’s Fellowship (USD 10,000.00) Fall 2023
Department of Astronomy, University of Maryland College Park

Chancellor’s Centennial Engineering Scholarship Spring 2023
University of Massachusetts Dartmouth

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| Graduate Research Award
<i>Department of Physics, University of Massachusetts Dartmouth</i> | Spring 2022 |
| FIP Distinguished Student Award (award + travel grant)
<i>American Physical Society</i> | Spring 2022 |
| DAP Student Travel Grant (USD 600.00)
<i>American Physical Society</i> | Spring 2022 |
| Gawad Direktor para sa Natatanging Bagong Guro
<i>National Institute of Physics, University of the Philippines Diliman</i> | December 2018 |
| <ul style="list-style-type: none"> · The award was given in recognition of the exemplary performance of a newly hired junior faculty of the institute. | |
| Gawad Direktor para sa Natatanging Discussion Teacher
<i>National Institute of Physics, University of the Philippines Diliman</i> | December 2018 |
| <ul style="list-style-type: none"> · This award is given in recognition of the exemplary performance of a junior faculty member as a discussion teacher for lecture classes offered by the institute. | |

PROFESSIONAL EXPERIENCE

4 years of teaching experience, ~ 6 years of independent research experience

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|---|-------------------------------------|
| Graduate Research Assistant
<i>Department of Astronomy, University of Maryland College Park</i> | 01/2024–present (2 years, 5 months) |
| <ul style="list-style-type: none"> · Leads a computational project that explores how magnetic fields evolve in galaxies through large-scale simulations using codes like ART, Arepo, and RAMSES | |
| Graduate Teaching Assistant
<i>Department of Astronomy, University of Maryland College Park</i> | 08–12/2023 (5 months) |
| <ul style="list-style-type: none"> · Teaching assistant for Prof. Lee Mundy's ASTR 101 (Fall 2023) · Led weekly discussions with a non-technical audience on foundations of astronomy and astrophysics · Supervised and graded laboratory work of a class with 20 students at a time | |
| Graduate Student Intern
<i>Los Alamos National Laboratory</i> | 06–08/2023 (3 months) |
| <ul style="list-style-type: none"> · Participant during the 2023 installment of the LANL Co-Design Summer School (est. 2011, http://lanl.github.io/cdss/) where we developed a performance portable code for thermonuclear supernovae | |
| Graduate Teaching Fellow
<i>Department of Physics, UMass Dartmouth</i> | 09/2022–05/2023 (9 months) |
| <ul style="list-style-type: none"> · Designed (as Instructor on Record) an introduction to astronomy course for a non-technical audience, with topics ranging from history, stellar evolution, galaxies, cosmology and indigenous astronomy. Graded exams and class output designed to synthesize lectures delivered in class. | |
| Graduate Research Assistant
<i>Department of Physics, UMass Dartmouth</i> | 05/2021–08/2022 (1 year, 3 months) |
| <ul style="list-style-type: none"> · Led computational projects on type Ia supernovae that explore (1) a first-principles turbulently-driven explosion mechanism in near-Chandrasekhar mass white dwarfs, (2) the magnetized evolution of white dwarf merger remnants. · Co-led a paper that illustrated how the isotope Ti-44 can be used to constrain the progenitor of a type Ia supernova event (<i>see Publications</i>) | |

Graduate Teaching Assistant

01–05/2021 (5 months)

Department of Physics, UMass Dartmouth

- Taught online recitation and laboratory courses in Classical Mechanics

Instructor 1

08/2018–12/2020 (2 years, 5 months)

National Institute of Physics, University of the Philippines Diliman

- Taught in-person and online lectures and laboratory classes in Classical Mechanics, Electromagnetism, Physical Electronics, and Computational Physics
- Designed exams and in-class exercises for introduction to classical mechanics and electromagnetism classes
- Co-organized events for the department, served as course group leader for electromagnetism labs

SERVICE

UMD GRAD-MAP Co-Lead

01/2024–08/2026

Department of Astronomy, University of Maryland College Park

- Co-authored a successful Do Good Institute Grant proposal (USD 6,000.00) with Yash Gursahani, where Mr Gursahani is the lead author
- Co-authored a successful Maryland Space Grant Consortium proposal (USD 28,340.00) with Yash Gursahani and Benedikt Diemer
- Prepared yearly budgets, materials for programs, and other logistics
- Organized Journal Clubs led by our summer scholars, provided feedback and moderated each session
- Developed our website at <https://www.umdgradmap.org/> using Jekyll
- Moderated and created content for our social media accounts on Instagram and LinkedIn
- Moderated professional skills workshops on how to give elevator pitches, coding in Python, creating documents with LaTeX and how to prepare and give a scientific talk
- Served as a research mentor to one student (Jin Young Kim, *now at UNC Chapel Hill*)

UMD GRAD-MAP Code Lab developer

06–11/2026

Department of Astronomy, University of Maryland College Park

- Develops open-source learning materials for GRAD-MAP Do Good Institute project “Code Lab,” with co-developers Sander Somers and Krish Patel, and project lead Yash Gursahani

Graduate Student Admissions Volunteer

02/2024 - present

Department of Astronomy, University of Maryland College Park

- Co-leads the 2027 prospective student visit planning committee with Sander Somers
- Planned and coordinated the Fall 2026 first-year graduate student orientation with Sander Somers and Keaton Donaghue, presented an introduction to campus and department resources available to graduate students
- Participated in prospective student interviews from 2024–2026

Graduate Student Council Representative

03/2026 - present

Department of Astronomy, University of Maryland College Park

- Represents 4th year graduate students in our department student body

INVITED PRESENTATIONS

National Insitute of Physics GANAP theory seminar

01/26/2026

*University of the Philippines Diliman, Quezon City, Philippines***University of Massachusetts Dartmouth Physics seminar**

09/12/2025

North Dartmouth, MA, U.S.A.

Harvard-Smithsonian Center for Astrophysics <i>Cambridge, MA, U.S.A.</i>	09/11/2025
1st Philippine Space Science and Astronomy Research Conference <i>Manila, Philippines (online)</i>	08/11/2025
Ateneo de Manila University LeaPs <i>Manila, Philippines (online)</i>	11/11/2023
Center for Theoretical Astrophysics seminar <i>Los Alamos National Laboratory</i>	06/15/2023
MIT Lunch Talk Series <i>Cambridge, MA, U.S.A.</i>	01/31/2022
University of Massachusetts Dartmouth Physics seminar <i>North Dartmouth, MA, U.S.A.</i>	02/04/2021

CONTRIBUTED PRESENTATIONS

RAMSES User Meeting 2026 (talk) <i>Yonsei University, Seoul, South Korea</i>	04/29/2026
Magnetic Fields and Cosmic Rays across Diverse Scales: What's Next? (poster) <i>Harvard-Smithsonian Center for Astrophysics, Cambridge, MA, U.S.A.</i>	09/10/2025
Supercomputing 2023 (poster) <i>Denver, CO, U.S.A.</i>	11/12–17/2023
Los Alamos Student Symposium (poster) <i>Los Alamos, NM, U.S.A.</i>	2023
American Astronomical Society Meeting 242 (poster) <i>Albuquerque, NM, U.S.A.</i>	06/07/2023
EuroWD22 Conference (talk) <i>Eberhard Karls Universität Tübingen, Tübingen, Germany</i>	08/15–19/2022
American Physical Society April meeting (talk) <i>New York City, NY, U.S.A.</i>	04/12/2022
38th Samahang Pisika ng Pilipinas Physics Conference (talk) <i>Manila, Philippines (online)</i>	10/19/2020

SCHOOLS ATTENDED

Burgers Program Summer School on Turbulence <i>University of Maryland Burgers Program for Fluid Dynamics</i>	06/03—07/2024
LANL Co-Design Summer School <i>Los Alamos National Laboratory, NM, U.S.A.</i>	05–08/2023
MESA Summer School <i>University of California Santa Barbara, CA, U.S.A.</i>	08/08–12/2022
NSF/APS-DPP GPAP Summer school on plasma physics for astrophysicists <i>Swarthmore College (online)</i>	06/07–11/2021
Deciphering Dark Matter: From Galaxies to the Universe <i>Institut Teknologi Bandung, Bandung, West Java, Indonesia (online)</i>	09/14–25/2020

SKILLS

Computer Languages Software & Tools	Python, Fortran, C, C++ <i>knowledgeable</i> in Julia and R ART (C, Kravstov et al), RAMSES (Fortran, Teyssier), Arepo (C, Springel), SWIFT (C, Schaller et al), FLASH (Fortran, Fryxell et al), MESA (Fortran, Paxton et al), yt (Python, Turk et al), SuperNu (Fortran, van Rossum, Wollaeger), Torch (Fortran, Timmes)
HPC Systems	TACC Stampede3, Stampede2, UMassD CARNIE, UMD Zataran, Astra

PRESS

“GRAD-MAP Students, Mentors ‘Learn From Each Other’ ”(August 20, 2025)

PROFESSIONAL REFERENCES

Prof. Benedikt Diemer

Adviser (PhD) and Associate Professor of Astronomy
diemer (at) umd (dot) edu
Department of Astronomy, UMaryland College Park

Dr. Hyun Lim

Staff Scientist, CAI-1 Applied Computer Science
hyunlim (at) lanl (dot) gov
Los Alamos National Laboratory

Prof. Bo Dong

Professor of Mathematics
bdong (at) umassd (dot) edu
Department of Mathematics, UMass Dartmouth

Prof. Robert Fisher

Adviser (MSc) and Professor of Physics
robert.fisher (at) umassd (dot) edu
Department of Physics, UMass Dartmouth

Prof. Ian Vega

Adviser (BSc and MSc) and Professor of Physics
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National Institute of Physics, UP Diliman

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